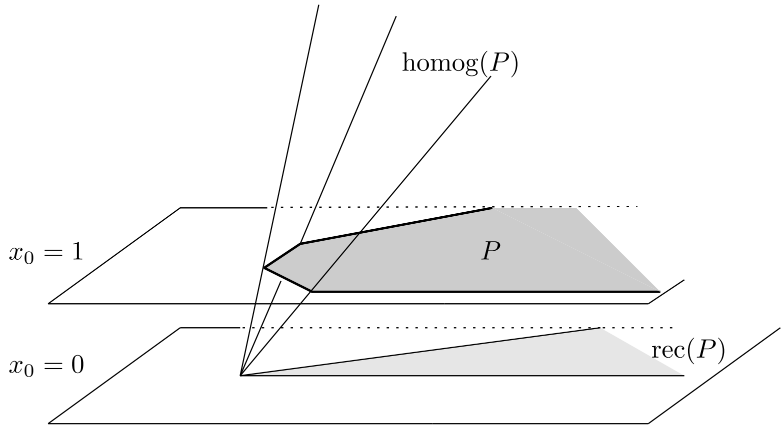
Again it is quite easy to see that for every convex set P, the homogenization $\mathrm{homog}(P)$ is a convex cone in $\mathbb{R}^{d+1}$. Furthermore, any P can be easily recovered from its homogenization (if we don't mess with the coordinate system) as

$$P = \{\boldsymbol{x} \in \mathbb{R}^d : \begin{pmatrix} 1 \\ \boldsymbol{x} \end{pmatrix} \in \mathrm{homog}(P)\}.$$



Proposition 1.14. *Let $P \subseteq \mathbb{R}^d$ be a convex set.*

(i) *If $P = P(A, \boldsymbol{z})$ is an $\mathcal{H}$-polyhedron, then its homogenization is also an $\mathcal{H}$-polyhedron:*

$$\mathrm{homog}(P) \;=\; P\Big(\begin{pmatrix} -1 & \mathbb{O} \\ -\boldsymbol{z} & A \end{pmatrix}, \begin{pmatrix} 0 \\ \mathbf{0} \end{pmatrix}\Big) = C(P).$$

(ii) *If $P = \mathrm{conv}(V) + \mathrm{cone}(Y)$ is a $\mathcal{V}$-polyhedron, then so is its homogenization:*

$$\mathrm{homog}(P) \;=\; \mathrm{cone}\begin{pmatrix} \mathbb{1} & \mathbb{O} \\ V & Y \end{pmatrix} = C(P).$$

Proof. This now follows from Proposition 1.12. □

1.6 Carathéodory's Theorem

The following proposition states two versions (linear and affine) of another basic tool, known as Carathéodory's theorem. We want to emphasize that in contrast to the Farkas lemma this is completely elementary and also computationally quite trivial. However, it can be successfully applied, for example, to sharpen the Farkas lemmas, as well as the main theorems and representation theorems for polytopes, cones, and polyhedra; see the next lecture.

Proposition 1.15 (Carathéodory's theorem).
Let $X \in \mathbb{R}^{d\times n}$ and $\boldsymbol{x} \in \mathbb{R}^d$.

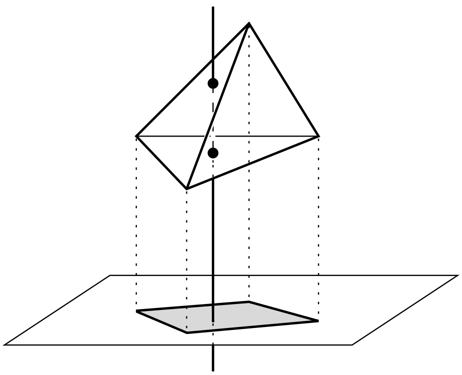
(i) *If $\boldsymbol{x} \in \operatorname{cone}(X)$, then $\boldsymbol{x} \in \operatorname{cone}(X')$ holds for a subset $X' \subseteq X$ of at most* $\operatorname{rank}(X) = \dim(\operatorname{cone}(X))$ *vectors in X.*

(ii) *If $\boldsymbol{x} \in \operatorname{conv}(X)$, then $\boldsymbol{x} \in \operatorname{conv}(X')$ holds for a subset $X' \subseteq X$ of at most* $\operatorname{rank}\begin{pmatrix} \mathbb{1} \\ X \end{pmatrix} = \dim(\operatorname{conv}(X)) + 1$ *vectors in X.*

We first describe the geometric idea (linear version). For this let $\operatorname{cone}(X)$ have dimension k; assume that $k' \geq k+1$ is the smallest number such that $\boldsymbol{x} \in \operatorname{cone}(X)$ can be represented as a positive sum k' vectors in X. (We obtain $k' \leq n$ from $\boldsymbol{x} \in \operatorname{cone}(X)$.)

The cone $\operatorname{cone}(X')$ spanned by such a set $X' \subseteq X$ of k' vectors in X can be interpreted as a projection of the positive orthant in $\mathbb{R}^{k'}$. Since k' is minimal, the point $\boldsymbol{x}$ lies in the image of the interior of the orthant, $\{\boldsymbol{t} \in \mathbb{R}^{k'} : \boldsymbol{t} > \mathbf{0}\}$. From $k' > k$ we get that the preimage of $\boldsymbol{x}$ under that projection is at least 1-dimensional. Thus the preimage contains the intersection of a line with the orthant $\{\boldsymbol{t} \in \mathbb{R}^{k'} : \boldsymbol{t} \geq \mathbf{0}\}$. Since the orthant does not contain a whole line, the preimage contains a boundary point of the orthant, and thus $\boldsymbol{x}$ can be represented as a conical combination of fewer than k' vectors.

Similarly (affine version), we consider the projection of a simplex to the polytope $\operatorname{conv}(X)$. If the image polytope has smaller dimension than the simplex, then any point of the polytope has as preimage the intersection of the simplex with a line. But the simplex does not contain a whole line, so the line must contain a boundary point of the simplex, which leads to a representation with fewer nonzero coefficients.



We will now give an algebraic proof.

Proof. For (i), without loss of generality we assume that X has full rank, $\operatorname{rank}(X) = d$, by passing to the linear hull of X. Now let $\boldsymbol{x} \in \operatorname{cone}(X)$ and let $\boldsymbol{x} = X\boldsymbol{t}$ with a vector $\boldsymbol{t} \geq \mathbf{0}$ of minimal support $\operatorname{supp}(\boldsymbol{t}) = \{i : t_i > 0\}$, that is, minimal number $|\{i : t_i > 0\}|$ of nonzero components. Now if $|\operatorname{supp}(\boldsymbol{t})| > d$, then $\{\boldsymbol{x}_i : t_i > 0\}$ is linearly dependent. This means that there is a linear dependence of the form $0 = \sum_{i=1}^n \lambda_i t_i \boldsymbol{x}_i$ with all $\lambda_i \neq 0$. By multiplying this with a nonzero $\alpha \in \mathbb{R}$ we may assume that $\lambda_i > 0$ for some $i \in \operatorname{supp}(\boldsymbol{t})$, and that $\max\{\lambda_i : t_i > 0\} = 1$. But then we get

$$\boldsymbol{x} \;=\; \sum_i t_i \boldsymbol{x}_i \;=\; \sum_i (1 - \lambda_i) t_i \boldsymbol{x}_i,$$

which is a representation with smaller support, contradicting the minimality of $\boldsymbol{t}$.

Now (ii) follows directly from

$$\boldsymbol{x} \in \operatorname{conv}(X) \Longleftrightarrow \begin{pmatrix} 1 \\ \boldsymbol{x} \end{pmatrix} \in \operatorname{cone}\begin{pmatrix} \mathbb{1} \\ X \end{pmatrix}. \qquad \square$$

Notes

The material of this lecture is classical — our discussion is inspired by Grötschel's treatment in [245].

We recommend Schrijver's book [484, Sect. 12.2] for more historical comments, as a superb guide to the historical sources, with references to the original papers by Fourier, Dines, and Motzkin, and also those by Minkowski, Weyl, Farkas, Carathéodory, and others.

The elimination method was developed by Motzkin in his 1936 doctoral thesis [414]. We quote from Dantzig & Eaves [175]:

> For years the method was referred to as the *Motzkin* Elimination Method. However, because of the odd grave-digging custom of looking for artifacts in long forgotten papers, it is now known as the *Fourier-Motzkin* Elimination Method and perhaps will eventually be known as the *Fourier-Dines-Motzkin* Elimination Method.

The Fourier-Motzkin elimination method is not only a theoretical tool — with some care it can also be used for computations. "In practice," however, one has to deal with an effect known as *combinatorial explosion*: every elimination step may transform m inequalities into up to $\lfloor m^2/4 \rfloor$ new ones, which means that after a few steps the number of inequalities in the system can increase dramatically. However, many of the inequalities we get by elimination are *redundant*: they can be deleted without changing the polyhedron that is described by the system. Thus it is important to eliminate redundant inequalities from the system, and in fact to detect many of them quickly, in order to keep the problem size and the computation times down.